NICRON 80mg Tablet

COMPOSITION

Gliclazide	80mg/tablet

PRODUCT DESCRIPTION

A white to off white round shaped tablet with double score line on one side and plain on other side.

PHARMACODYNAMICS

Gliclazide is a second generation sulphonylurea oral hypoglycaemic agent. Gliclazide reduces blood glucose levels by stimulating the insulin secretion βcells in the islets of Langerhans. Unstimulated and stimulated insulin secretion from pancreatic β-cells are increased following administration of Gliclazide. This occurs via the binding of Gliclazide to specific receptor on pancreatic β-cells which results in a decrease in potassium efflux and causes depolarisation of the cell. Subsequently, calcium channels open, leading to an increase in intracellular calcium and induction of insulin release. In addition, Gliclazide increases the sensitivity of β-cells to glucose. A significant increase in insulin response is seen in response to stimulation induced by a meal or glucose.

Haemovascular properties: Gliclazide decreases microthrombosis by two mechanisms which may be involved in the complications of diabetes: A partial inhibition of platelet aggregation and adhesion with a decrease in the markers of platelet activation (beta-thromboglobulin, thromboxane B2), an action on the vascular endothelium fibrinolytic activity with an increase in tPA activity.

PHARMACOKINETICS

Absorption: Gliclazide is rapidly absorbed from the gastrointestinal tract and maximum blood concentrations are attained between the 11th and the 14th hour.
Distribution: In humans, binding to proteins is 94.2%. Metabolism is extensive and all the metabolites are devoid of hypoglycaemic activity.
Elimination: As the apparent terminal elimination half-life for Gliclazide is 20 hours in humans, the drug can be administered in two daily doses.
Elimination is mainly in the urine: less than 1% of the dose ingested is found unchanged in the urine.

INDICATION

Non-insulin dependent diabetes, in association with an adapted diet, in cases where dietary measures alone provide inadequate control blood glucose levels.

RECOMMENDED DOSAGE
RESTRICTED TO ADULTS.

As with any hypoglycaemic agent, the dosages should be adjusted to the particular circumstances. In the event of a transitory loss of blood glucose control in a patient where good control is usually achieved by diet, it may be sufficient to administer this product for a short period.

<u>Patients under the age of 65 Initial dose:</u> The recommended initial dose is one tablet per day.
Dosage steps: Adjustments in posology are usually made in steps of one tablet at a time, depending on the response in blood glucose. At least 14 days should separate successive steps in the dose.

<u>Maintenance treatment:</u> The posology can vary from 1 to 3, or rarely 4, tablets per day. The standard dosage is 2 tablets a day, as 2 daily doses.

Patients at particular risk

Patients over the age of 65: Begin the treatment with a half-tablet once a day. The dose can be progressively increased until the patient's blood glucose is satisfactorily controlled, while keeping to steps of at least 14 days between successive levels, and with close monitoring of blood glucose.

Other patients at particular risk: In patients who are malnourished or showing markedly poor general state of health, or whose calorie intake is irregular, or whose kidney or liver function is impaired, treatment should be begun at the lowest dose. The stepwise increases in dosage must be scrupulously adhered to in order to avoid a hypoglycaemic reaction

Patients treated with other oral hypoglycaemic agents: As with any sulfonylurea, this medicinal product can be used as follow-on from another antidiabetic drug, without any transitional period being needed. When patients change to this medication from a sulfonylurea with a longer half-life (such as chlorpropamide), they must be closely monitored (for several weeks). This is to avoid the possible occurrence of hypoglycaemia due to an overlapping of effects from the two treatments.

Paediatric population: The safety and efficacy of Gliclazide 80 mg in children and adolescents have not been established. No data are available.

ROUTE OF ADMINISTRATION

Oral

CONTRAINDICATIONS

Hypersensitivity to Gliclazide, sulfonylureas or sulphonamides. Type 1 diabetes, diabetic pre-coma and coma, diabetic keto-acidosis. Severe renal and hepatic impairment. Concomitant use with miconazole Pregnancy and lactation.

WARNING AND PRECAUTIONS

Hypoglycaemia

This medication should be prescribed only to patients who are able to feed themselves regularly (including having breakfast).

It is important to consume carbohydrates regularly, in view of the increased risk of hypoglycaemia if meals are taken late, if food intake is inadequate or if there is an imbalance in carbohydrates.

Hypoglycaemia is all the more likely to occur when the patient's diet is low in calories, following major or prolonged exercise, following ingestion of alcohol or when a combination of hypoglycaemic agents is being administered.

Hypoglycaemia can occur following administration of sulfonylureas. Some episodes can be severe and prolonged.

Admission to hospital may then be necessary, and glucose administration may need to be carried out for several days.

To reduce the risk of hypoglycaemia, the patient must be carefully selected, the posology must be carefully matched to the patient, and the patient must be given appropriate information. Factors which increase the risk of hypoglycaemia:

- Patient refuses or is unable to co-operate (particularly in elderly subjects),
- Malnutrition, irregular mealtimes, skipping meals, periods of fasting or dietary changes,
- Imbalance between physical exercise and carbohydrate intake,
- Renal insufficiency,
- Severe hepatic insufficiency,
- Overdose of NICRON,

- Certain endocrine disorders: thyroid disorders, hypopituitarism and adrenal insufficiency,
- Concomitant administration of certain other medicines.

Renal and hepatic insufficiency
The pharmacokinetics and/or pharmacodynamics of Gliclazide may be altered in patients with hepatic insufficiency or severe renal failure. If hypoglycaemia occurs in such patients, it can be prolonged, so appropriate management should be initiated.

Information for the patient
The risks of hypoglycaemia, together with its symptoms, treatment, and conditions that predispose to its development, should be explained to patients and to family members. They must be informed of the importance of keeping to an appropriate diet, undertaking regular physical exercise, and of regular monitoring of blood glucose levels.

Blood sugar imbalance
Blood glucose control in a patient receiving oral antidiabetic treatment may be affected by any of the following: St. John's Wort (Hypericum perforatum) preparations, fever, injury, infection or surgery.
In some cases, it may be necessary to administer insulin.
The hypoglycaemic efficacy of any oral antidiabetic agent, including Gliclazide, is attenuated over time in many patients: this may be due to progression in the severity of the diabetes or to a reduced response to treatment. This phenomenon is known as secondary failure which is distinct from primary failure, when an active substance is ineffective as first-line treatment. Adequate dose adjustment and dietary compliance should be considered before classifying the patient as secondary failure.

Dysglycaemia
Disturbances in blood glucose, including hypoglycaemia and hyperglycaemia have been reported in diabetic patients receiving concomitant treatment with fluoroquinolones, especially in elderly patients. Indeed, careful monitoring of blood glucose is recommended in all patients receiving at the same time NICRON 80 mg and a fluoroquinolone.

Laboratory tests
Measurement of glycated haemoglobin levels (or fasting venous plasma glucose) is recommended in assessing blood glucose control. Blood glucose self-monitoring may also be useful. Treatment of patients with G6PD-deficiency (glucose-6-phosphate dehydrogenase deficiency) with sulfonylurea agents can lead to haemolytic anaemia. Since Gliclazide belongs to the chemical class of sulfonylurea drugs, caution should be used in patients with G6PD-deficiency and a non-sulfonylurea alternative should be considered.

Excipients
This medicinal product contains lactose. Patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose-galactose malabsorption should not take this medicine.

Effect on ability to drive and use machines
NICRON 80 mg has no or negligible influence on the ability to drive and use machines. However, patients should be made aware of the symptoms of hypoglycaemia and should be careful if driving or operating machinery, especially at the beginning of treatment.

DRUG INTERACTIONS
Increased hypoglycaemic effect with phenylbutazone. May decrease metabolism with chloramphenicol. May enhance effect of anticoagulant agents (e.g. warfarin). May enhance hypoglycaemic effect with

other antidiabetic agents (e.g. insulin, metformin, glucagon-like peptide-1 receptor agonists, acarbose, thiazolidinediones, dipeptidyl peptidase-4 inhibitors), ß-blockers, fluconazole, salicylates, ACE inhibitors (e.g. captopril, enalapril), H_2-receptor blockers, MAOIs, sulfonamides, quinolone antibiotics, clarithromycin, and other NSAIDs. Increased blood glucose levels with chlorpromazine (at high doses), ß-agonists (e.g. salbutamol, ritodrine), tetracosactrin, glucocorticoids (with possible ketosis), and danazol. **Potentially Fatal:** Increased hypoglycaemic effect and possible onset of hypoglycaemia, or coma with miconazole.

PREGNANCY AND LACTATION

Use in pregnancy: Gliclazide is contraindicated in pregnancy.

Use in lactation: Gliclazide is secreted in breast milk and patients taking gliclazide should not breast feed.

ADVERSE EFFECTS

Hypoglycaemia

The most frequent adverse reaction with Gliclazide is hypoglycaemia.

As for other sulfonylureas, treatment with NICRON 80mg can cause hypoglycaemia, in particular if mealtimes are irregular and, if meals are skipped.

Possible symptoms of hypoglycaemia are: headache, intense hunger, nausea, vomiting, lassitude, sleep disorders, agitation, aggression, poor concentration, reduced awareness and slowed reactions, depression, confusion, visual and speech disorders, aphasia, tremor, paresis, sensory disorders, dizziness, feeling of powerlessness, loss of self-control, delirium, convulsions, shallow respiration, bradycardia, drowsiness and loss of consciousness, possibly resulting in coma and lethal outcome.

In addition, signs of adrenergic counter-regulation may be observed: sweating, clammy skin, anxiety, tachycardia, hypertension, palpitations, angina pectoris and cardiac arrhythmia.

Usually, symptoms disappear after intake of carbohydrates (sugar). However, artificial sweeteners have no effect. Experience with other sulphonylureas shows that hypoglycaemia can recur even when measures prove effective initially.

If a hypoglycaemic episode is severe or prolonged, and even if it is temporarily controlled by intake of sugar, immediate medical treatment or even hospitalisation are required.

Other undesirable effects

Gastrointestinal disturbances, including abdominal pain, nausea, vomiting dyspepsia, diarrhoea, and constipation have been reported: if these should occur they can be avoided or minimised if gliclazide is taken with a meal or by splitting the doses.

The following undesirable effects have been more rarely reported:

- *Skin and subcutaneous tissue disorders:* rash, pruritus, urticaria, angioedema, erythema, maculopapular rash, bullous reactions (such as Stevens-Johnson syndrome and toxic epidermal necrolysis) and exceptionally, drug rash with eosinophilia and systemic symptoms (DRESS).
- *Blood and lymphatic system disorders:* changes in haematology are rare. They may include anaemia, leucopenia, thrombocytopenia, granulocytopenia. These are in general reversible upon discontinuation of medication.
- *Hepato-biliary disorders*: raised hepatic enzyme levels (AST, ALT, alkaline phosphatase), hepatitis (isolated reports). Discontinue treatment if cholestatic jaundice appears. These symptoms usually disappear after discontinuation of treatment.
- *Eye disorders:* transient visual disturbances may occur especially on initiation of treatment due to changes in blood glucose levels.

Class attribution effects

As for other sulphonylureas, the following adverse events have been observed: cases of erythrocytopenia, agranulocytosis, haemolytic anaemia, pancytopenia, allergic vasculitis, hyponatraemia, elevated liver enzyme levels and even impairment of liver function (e.g. with cholestasis and jaundice) and hepatitis which regressed after withdrawal of the sulfonylurea or led to life-threatening liver failure in isolated cases.

OVERDOSE AND TREATMENT

Hypoglycaemic reactions with or without coma, convulsions or other neurological disorders.
Management: Consider carbohydrate intake, or modification of diet and dose adjustment.

STORAGE CONDITIONS

Store below 30°C.
Protect from light.

PRESENTATION

PVC/Aluminium blister of 10 tablets.
Pack Sizes: 60's, 100's, 120's, 500's and 1000's tablets

MANUFACTURED AND PRODUCT REGISTRATION HOLDER

Noripharma Sdn Bhd. 200701034604 (792633-A)
Lot 5030 Jalan Teratai,
5½ Mile Off Jalan Meru,
41050 Klang, Selangor, Malaysia.

DATE OF REVISION

May 2026